

Yuanlong WEI

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Dezhou City, Shandong Province, China

EDUCATION

Harbin Institute of Technology (C9 / 985 / Double First-Class)

Harbin, Heilongjiang, China

Master of Engineering in Traffic and Transportation Engineering

Sep. 2022 – Jun. 2025

Thesis: *Research on Intelligent Passivity-Based Control Techniques for Chained Integrator Dynamic Systems (CAV platoon as verification platform)*

Jilin University (985 / Double First-Class)

Changchun, Jilin, China

Bachelor of Engineering in Traffic Engineering

Sep. 2018 – Jun. 2022

GPA: 3.62/4.0 (Top 11%)

MAIN ACADEMIC PROJECTS

National Natural Science Foundation No. 62303134

Jan. 2024 – Dec. 2026

Key Project Participant / Supervisor: Associate Professor Yunfei Yin

- Took charge of 14 rounds of grant proposal drafting and language polishing, covering research motivation, literature review, theoretical framework and technical schemes.
- Mastered rigorous academic logic and precise technical expression for formal journal and project writing.
- Developed a model-free SMC algorithm for automated vehicles, generalizable to robot control systems.

HIT “0-1 Original Exploration” Idea Fund No. HIT.DZJJ.2023084

Nov. 2023 – Nov. 2024

Project Leader / Supervisor: Associate Professor Yunfei Yin

- Proposed VO-IDA, a novel passivity-based control framework for general chained-integrator dynamics, validated on connected vehicle platoon systems.
- Completed full theoretical mathematical derivation (500+ pages over 6 months).
- Built proficiency in LaTeX manuscript writing and vector graphic design with Inkscape.

PUBLICATIONS

Journal Papers

1. Yin, Y. (Supervisor), **Wei, Y. (1st co-author)**, Dong, Z., Xue, M., Vazquez, S., Wu, L.
V2V-Based Cooperative Control of Heterogeneous CAV Platoons: An Intelligent VO-IDA Approach
IEEE Internet of Things Journal, **2024**, 11(22): 36257 | Published
2. **Wei, Y. (1st author)**, Yin, Y., Jiang, W., Dong, Z., Vazquez, S., Xu, Z.
Robust NN Adaptive VO-IDA Control Against Uncertain Dynamics for CAV Platoons
IEEE Transactions on Industrial Informatics | Under Review.
3. Yin, Y., Xu, Z., **Wei, Y. (3rd author)**, Jiang, W., Yao, D., Dong, Z.
Model-Free Sliding Mode Control for Path Following in Autonomous Vehicle
Journal of the Franklin Institute, **2025**, 362(12): 107806 | Published
4. Yin, Y., Jiang, W., **Wei, Y. (3rd author)**, Shangguan, L., Xu, Z., Zong, Z.
Fusion Estimation of Tire Road Adhesion Coefficient Based on Tire-Icy Road Interaction Simulation and Dynamic Information under Icy Road Driving Conditions
IEEE Transactions on Automation Science and Engineering | Under Review

Conference Papers

5. Xu, Z., Yin, Y., **Wei, Y. (3rd author)**, Jiang, W.
Improved Model-Free Adaptive Control and Line-of-Sight Guidance for Autonomous Vehicle Path Following
IEEE International Conference on Industrial Technology (ICIT 2025).
6. Zhu, W., Yin, Y., Xu, Z., **Wei, Y. (4th author)**, Zeng, S.
Multi-objective Cooperative Control of Vehicle Platoon Based on Fuzzy MPC
IEEE International Conference on Industrial Technology (ICIT 2025).

ACADEMIC SKILLS

Core Control & Stability Theory

Passivity-Based Control (IDA-PBC), Sliding Mode Control (SMC), Neural Network Adaptive Control, MPC, Backstepping, Fuzzy Control, ADRC, Linear System Theory, Lyapunov Stability Analysis

AI, Learning & Simulation Implementation

- Learning Algorithms: Deep Learning, Reinforcement Learning, Genetic Algorithms
- Coding and Simulation Tools: MATLAB + Simulink (Proficient), Python + PyTorch, C, Excel VBA

Academic Writing & Visualization Tools (Proficient)

LaTeX, Markdown, Zotero / Bookends (BibTeX), PowerPoint, Inkscape, Photoshop, MindNode, AutoCAD

Supplementary Transportation Domain Skills

CarSim-Simulink Co-Simulation, VISSIM, TransCAD

Language Proficiency

English (Fluent), Chinese (Native)

AWARDS & SCHOLARSHIPS

Top-Class Academic Scholarship, Harbin Institute of Technology (2023)

Second-Class Academic Scholarship, Harbin Institute of Technology (2023, 2024)

Outstanding Undergraduate Thesis, Jilin University (2022)

National Third Prize, Road Safety Innovation Competition (2020)

College Outstanding Student, Jilin University (2019-2022, 4 times)

Second-Class Scholarship, Jilin University (2019-2022, 4 times)

Social Work Award, Jilin University (2019)